

EXPERIMENT-2- Monoalphabetic Substitution Cipher

AIM: To write a C program to implement the Monoalphabetic Substitution Cipher, where each plaintext alphabet is replaced with a unique corresponding ciphertext alphabet using a substitution key.

CODE:

```
#include <stdio.h>
```

```
#include <ctype.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```
    char text[100], key[27] = "QWERTYUIOPASDFGHJKLZXCVBNM";
```

```
    int i;
```

```
    printf("Enter plaintext: ");
```

```
    fgets(text, sizeof(text), stdin);
```

```
    for(i = 0; text[i] != '\0'; i++)
```

```
    {
```

```
        if(isupper(text[i]))
```

```
            text[i] = key[text[i] - 'A'];
```

```
        else if(islower(text[i]))
```

```
            text[i] = tolower(key[text[i] - 'a']);
```

```
}
```

```
printf("Encrypted text: %s", text);
```

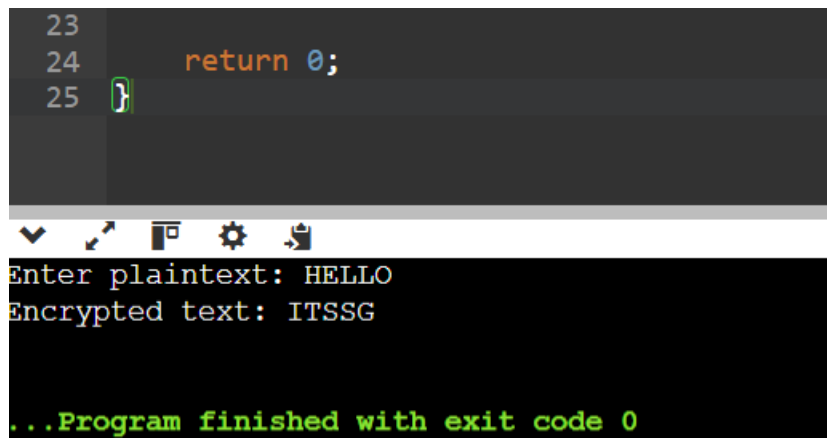
```
return 0;
```

```
}
```

INPUT:HELLO

KEY: 3

OUTPUT:



The image shows a code editor window with a dark background. The editor displays three lines of code: line 23 is empty, line 24 contains `return 0;`, and line 25 contains a closing curly brace `}`. Below the editor is a terminal window with a black background. The terminal shows the following text: `Enter plaintext: HELLO`, `Encrypted text: ITSSG`, and `...Program finished with exit code 0`.

```
23  
24     return 0;  
25 }
```

```
Enter plaintext: HELLO  
Encrypted text: ITSSG  
...Program finished with exit code 0
```